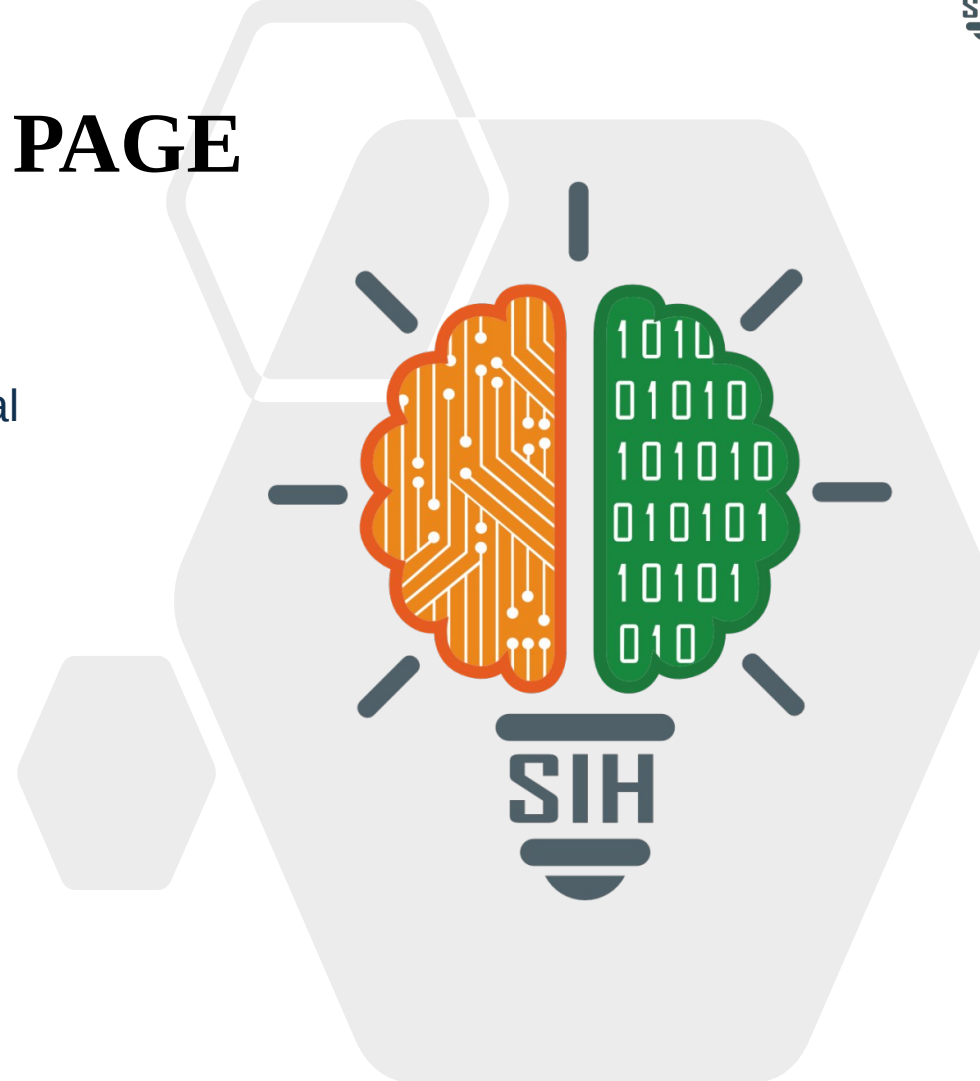




TITLE PAGE

- **Problem Statement ID** – SIH26167
- **Problem Statement Title** – SatQuery AI: An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries
- **Theme** – Space Technology
- **PS Category** – Software
- **Team ID** – *[enter Team ID from SIH portal]*
- **Team Name** – Saverra



Ask the Earth in plain English. Get evidence-backed answers.

Proposed Solution: SatQuery AI is an agentic vision-language assistant that turns a plain-English question into the right satellite analysis, across optical, multispectral and SAR imagery.

Detailed Solution

- **Chat + map UI:** choose region and dates, ask in plain English
- **LLM agent** identifies the task: VQA, captioning, land cover, change
- **Routes** to the right model for optical, multispectral or SAR
- **Returns** answer + map overlay + evidence + confidence
- **Live data** from ISRO Bhoonidhi and Sentinel

How It Addresses the Problem

- **No expertise needed:** no sensors, bands or GIS tools to learn
- **One platform** replaces many single-task tools
- **SAR works through** monsoon clouds when optical fails
- **Verifiable:** masks and confidence, not a black box
- **Opens EO** to students, researchers, field officers

Innovation: SatQuery AI vs Existing Approaches

	GIS / task-specific	RS VLMs (e.g. GeoChat)	SatQuery AI
Plain-language queries	✗	✓	✓
Many RS tasks, one system	✗	~	✓
Optical, multispectral, SAR	~	✗	✓
Multi-date change Q&A	~	✗	✓
Auto model/tool selection	✗	✗	✓
Evidence + confidence	~	✗	✓

✓ supported ~ partial ✗ not supported (typical, based on published capabilities)

EXAMPLE: USER ASKS

"How much has the flood extent near Guwahati changed between 1 June and 15 July 2024?"

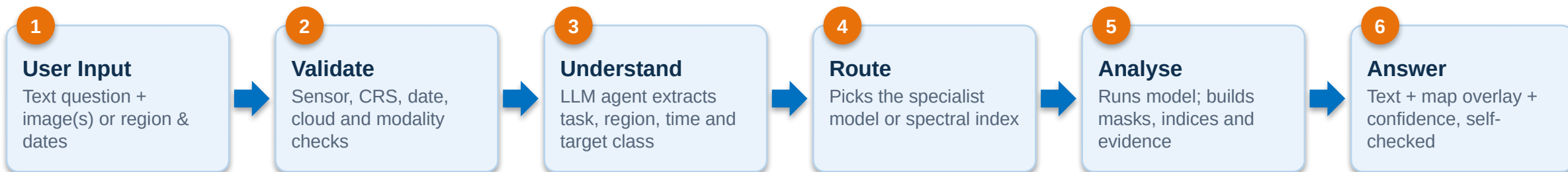
AGENT PLANS

Fetch Sentinel-1 SAR + Sentinel-2 → water segmentation
→ multi-date change computation

SATQUERY RETURNS

Answer in km², change map overlay and confidence score

METHODOLOGY & PROCESS FLOW



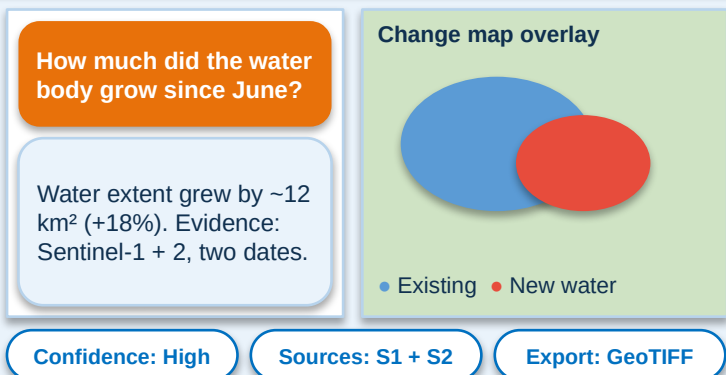
TOOLBOX THE AGENT ROUTES TO



Technologies Used

Frontend: React / Next.js, MapLibre map viewer
Backend: Python, FastAPI, Docker
AI / ML: PyTorch, Hugging Face, LoRA / QLoRA
Models: GeoChat-class RS VLM + LLM planner
Geospatial: Rasterio, GeoPandas, Xarray
Data: Bhoonidhi API, Sentinel-1/2

Interface Concept (illustrative)



Evaluation & Trust

- **Benchmarks:** RSVQA, VRSBench, CDVQA, BigEarthNet.txt
- **Targets:** ≥90% correct tool routing on a curated test set; answers in under 30 s
- **Guardrails:** evidence maps, confidence score, abstain when unsure
- **Review:** expert check of sample answers on Indian scenes

Why It Is Feasible

- ✓ **Open benchmarks** cover every task: RSVQA, VRSBench, CDVQA, BigEarthNet.txt
- ✓ **Open-source RS VLMs** (e.g. GeoChat) to build on, so no training from scratch
- ✓ **LoRA / QLoRA** fine-tuning fits on a single GPU
- ✓ **Free imagery:** Sentinel-1/2 open data plus registered Bhoonidhi access
- ✓ **Modular design:** each tool works alone, so the system degrades gracefully

Potential Challenge

Large scenes, slow response

Optical and SAR behave very differently

GPU and compute limits

Confident but wrong answers

Bhoonidhi access and API limits

Mitigation Strategy

Tile and crop to the region of interest, cloud-optimised GeoTIFFs, result caching

Separate specialist models per modality; the agent fuses their outputs at the reasoning level

LoRA / QLoRA and quantised small models; start with data subsets, scale after the baseline works

Validate inputs, show evidence maps and confidence, abstain when unsure; benchmark before integrating

Prototype on open Sentinel data first; a pluggable data connector lets Bhoonidhi drop in

IMPLEMENTATION ROADMAP

Phase 1 · Baseline

Fine-tuned RS VLM answers questions on Sentinel scenes

Phase 2 · Specialists

Change, SAR and land-cover tools added

Phase 3 · Agent + Data

Router plans tasks; Bhoonidhi connector live

Phase 4 · Evaluate + Demo

Benchmarks, guardrails and a polished UI

TARGET AUDIENCE

Students & researchers

Disaster-response teams

Agriculture & forestry
officers

Urban & water planners

NGOs & citizens

Social Impact

- **Democratises Earth observation:** no GIS or remote-sensing expertise needed
- **Faster situational awareness** for floods, landslides and other disasters
- **Builds EO skills** in classrooms and research labs
- **Better decisions** by local officers and NGOs

Economic Impact

- **Design target:** minutes instead of hours or days of manual analysis
- **One platform** replaces several single-task tools
- **More value from national data:** ISRO Bhoonidhi and open Sentinel archives
- **Lower entry cost** for start-ups building EO applications

Environmental Impact

- **Routine monitoring** of forests, water bodies, crops and urban growth
- **Change-aware answers** quantify loss or gain, e.g. area in km²
- **SAR support** keeps monitoring possible through monsoon cloud cover
- **Supports SDG 11, 13 and 15** (cities, climate action, life on land)

Vision: every student, scientist and field officer can ask the Earth a question and get a trustworthy answer, powered by India's own Earth-observation data.

Datasets & Benchmarks

BigEarthNet.txt (2026)

Herzog, Adler et al. 464,044 Sentinel-1/2 pairs, 9.6M text annotations

[arXiv:2603.29630](#) · [txt.bigearth.net](#)

VRSBench (NeurIPS 2024)

Li, Ding, Elhoseiny. 29,614 images, 123K+ question-answer pairs

[arXiv:2406.12384](#) · [vrsbench.github.io](#)

RSVQA (IEEE TGRS 2020)

Lobry, Marcos, Murray, Tuia. Visual question answering for remote sensing

[arXiv:2003.07333](#)

CDVQA (IEEE TGRS 2022)

Yuan, Mou, Xiong, Zhu. Change-detection VQA, 2,968 image pairs

[arXiv:2112.06343](#)

Models & Methods

GeoChat (CVPR 2024)

Kuckreja et al. Remote-sensing VLM built on LLaVA-1.5 with LoRA

[arXiv:2311.15826](#)

LLaVA (NeurIPS 2023)

Liu et al. Visual instruction tuning

[arXiv:2304.08485](#)

LoRA (ICLR 2022)

Hu et al. Low-rank adaptation of large language models

[arXiv:2106.09685](#)

ReAct (ICLR 2023)

Yao et al. Reasoning + acting, the basis for tool-using agents

[arXiv:2210.03629](#)

Data Platforms & Problem Statement

Bhoonidhi

NRSC / ISRO Earth Observation data portal

[bhoonidhi.nrsc.gov.in](#)

Copernicus Data Space

Sentinel-1 (SAR) and Sentinel-2 (multispectral) open data

[dataspace.copernicus.eu](#)

Open-source stack

PyTorch, Hugging Face Transformers + PEFT, FastAPI, Rasterio, GeoPandas, Xarray

SIH26167: SatQuery AI

Problem statement by Indian Space Research Organisation (ISRO), Space Technology theme